

Pediatric Aspects in Young Rock Climbers

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Isabelle Schöffl and Volker Schöffl



Luc Schöffl in Halfway Log Dump, Ontario, Canada, Photo Michael Simon

I. Schöffl (✉)

Department of Pediatric Cardiology,
Friedrich-Alexander Universität Erlangen-Nürnberg,
Erlangen, Bayern, Germany

V. Schöffl

School of Clinical and Applied Sciences, Leeds
Beckett University, Leeds, UK

Department of Orthopedic and Trauma Surgery,
Center of Sportsmedicine, Klinikum Bamberg,
Bamberg, Germany

Section of Wilderness Medicine, Department of
Emergency Medicine, University of Colorado School
of Medicine, Denver, CO, USA

Department of Orthopedic and Trauma Surgery,
Friedrich-Alexander Universität Erlangen-Nürnberg,
Erlangen, Germany

14.1 Significance of Climbing for the Early Child Development

Exercise plays a central role for the physical and mental development of children. Climbing represents an exceptionally adequate exercise modality as climbing promotes several developments at once.

First of all, climbing combines endurance with power [1]. On the one hand, climbing therefore influences the maximal functional capacity in children (VO_2peak), a parameter, which has proven to be the best predictor for overall mortality and morbidity [2]. On the other hand, the fact that climbing is a sport in which the whole body is concerned, no muscle groups are at a disadvantage. However, muscular disbalances need to be kept in mind (Fig. 14.1).

In addition to its power and endurance characteristics, climbing demands a high amount of technical skills, balance, and coordination [3]. Especially, the motoric and cognitive promotion represents a distinctiveness which adds many benefits to a child's development. Interestingly, children know naturally how to move in the vertical environment. A lot of technical skills, like keeping one's central weight below the holding hand or how to turn one's body as to minimize the weight having to be supported, have to be laboriously learned by grown-ups, whereas children move correctly right from the start without ever being instructed. The movement engrams being

learned during climbing can then improve locomotion in daily life. In a study investigating the effects of a climbing intervention in children with infantile cerebral palsy was able to show that children improved their general functional capacity within 3 weeks [4].

Finally the mental and psychological aspects of climbing need to be discussed. Climbing demands a high amount of responsibility. The ability to judge real dangers and differentiate objective danger from irrational fear from falls is important and needs to be learned by the athlete [5] (Fig. 14.2). Whereas climbing has been deemed safe in several studies [6–8], certain precautions are necessary, and children need to be made aware of the objective dangers of the sport. Route-reading skills and a high demand on concentration [9] are a further advantage for the mental development of children.

Overall, climbing needs to adapt to the circumstances and respective capabilities of each child. The physical-mental development of small children often develops in large steps rather than



Fig. 14.1 Climbing advances the whole musculature



Fig. 14.2 Full concentration and responsibility already at an early age

in a continuous pattern. Physical, mental, and educational changes have a greater impact on the behavior and competence of children than usually observed in adults. In a sport, where a high amount of individual responsibility, self-assessment, and motoric capabilities is important, special attention must be paid to the development of each individual.

14.2 Specifics of Climbing with Children

The most important thing when climbing with children is safety. When climbing, the athlete places his/her life in the hands of another person, namely, the one holding the rope at the other end. This responsibility represents a special challenge for children but also teaches them a better comprehension of reality. Climbing, thus is the exact contrary of computer games, where actions performed in the game world lead to no comprehensible consequences for the child [10]. In order to learn from the climbing experience, children need to be able to assess themselves correctly and deduce the correct decisions. They need to learn to judge risks and act accordingly. This responsibility needs to be introduced slowly and according to their respective skills which need to be reevaluated constantly. Even skills the child had seemingly already acquired can change over the course of development and deteriorate. As an example, most children can understand that they shouldn't be playing underneath bouldering adults around the age of 6 years. However, when surrounded by peers and in a stressful or exciting situation, the importance of such rules is easily forgotten. However, if the child is slowly introduced to personal responsibility, a heightened self-esteem can be the result (Fig. 14.3).

The age, at which children can start belaying others has declined with the use of semi-automatic belaying devices. Under surveillance of adults with an added belay device, an 8-year-old child can, when adequately developed, belay another child of comparable weight in "top-rope." Lead-belaying of a weight-comparable peer can be performed, dependent on mental



Fig. 14.3 Even children can take responsibility, but it is essential to follow safety guidelines. Responsible "spotting" and crash pads are necessary

development, at the age of 12 [11]. Again, constant reevaluation of the skills of the concerned child is essential, always bearing in mind that regression of concentration can occur at any time.

With respect to the need for movement, we need to differentiate between children (up to 14 years of age) and adolescents (14–18 years of age). Especially up to the 12th birthday, body perception and movement knowledge should be in the center of the physical performance. Sometimes, it may be necessary to limit the movement compulsion in order to prevent one-sided strain or overuse syndromes. However, in most cases, children rarely perform exercises that cause pain or overuse. Physically motivated adolescents are different. Early disbalances and muscular contractions need to be observed early on and treated (s. below). It is difficult to give overall recommendations for certain age-groups as age does not always conform with development. Even though a team may consist of the same age-groups, their individual developments could differ over a wide range. This

is even more pronounced in mixed teams as girls tend to be ahead of boys in their mental and physical development by one and a half to 2 years. However, even within the same gender, there are substantial differences: a 15-year-old boy can still be on the developmental stage of puberty with open growth-plates and a body of a child with subsequent coordination deficits as a consequence of “unharmonic growth.” Another adolescent of this age may be already bodily mature and experience none of these difficulties. The maturity can be determined using the skeletal age (X-ray of the hand for determining biological age). If there is concern regarding developmental delay, endocrinological workup may be necessary. As restrictive energy intake is a cause for developmental delay, a proper workup with respect to the diet and mental status of the athlete is essential. A good solution for following the development is keeping growth, weight, and BMI curves of each athlete over time.

14.3 Training with Children and Adolescents

Main principal when training with children and adolescents is a high fun aspect and keeping the training age-appropriate. There are no concerns about cardiopulmonary overuse to date [12]. Even though, the cardiopulmonary function of each child/adolescent needs to be checked at regular intervals (s. Chap. 19, Sportsmedical Supervision of Climbers), an optimal climbing-specific training sets the necessary foundations at the right moments. A sport-specific power-training should be postponed until the skeletal growth is completed and climbing-specific coordination is already well developed. Sport-specific power training with high intensities, like campus boarding, should be avoided when working with children or adolescents [6]. As a consequence of a very high strength to body-weight ratio, such training is unnecessary and highly questionable for the athlete’s health. A study investigating the correlation between high-intensity campus boarding and the development of early arthrosis in the finger joints was able to prove movement con-

straints in finger extension [13]. Often, adolescents try to mimic the methods applied by high-level adult athletes, not knowing about the possible consequences. Even though high-level athletes may be able to tolerate such training modalities, they could be detrimental to the undeveloped skeletal system of adolescents. During childhood and adolescence, the main focus in training should therefore lie on learning motoric skills and improving movement coordination. When setting routes, the endurance aspect should be more pronounced than the ability to hold on to extremely small holds. The sensitive areas are the growth plates, the insertions of the tendons on the bone, and the relatively soft joint cartilage. Therefore, the most common injuries are epiphyseal fractures and disturbances [14], especially at or around of the time of the pubertal growth spurt [15]. This injury is explained in detail in the chapter regarding finger injuries and was lately found more often in adolescents training for speed climbing [16]. In rare cases, we have observed osteonecrosis of other bones, such as the capitulum humeri (M. Panner) (Fig. 14.4.) or the



Fig. 14.4 Osteonecrosis of the capitulum humeri—M. Panner in a 12-year-old climber

metacarpalia. In these cases, it is difficult to be certain whether climbing itself was responsible for these rare cases or whether climbing only helped demask the pathology, and the cause is probably multifactorial. Special attention needs to be paid on the proper development of the spine in order to detect early rounding in the sense of a climbers back [17] or scoliosis. Functional “scapula alatae” can be observed regularly and most commonly correct themselves, whereas special caution is needed with respect to gliding vertebrae.

Another point that needs to be discussed when working with children in climbing is the proper choice of shoes. Even though adults have the tendency to choose especially small and tight shoes, this does not work with children. Climbing shoes should never be too small or too tight as they could restrict bone growth (Figs. 14.5 and 14.6).

In addition to that, a climbing backpack has to be adapted to the child’s weight and height. Even

though the school bag of many children may be exceptionally heavy and large, this does not mean they can carry the same amount in an alpine expedition. The weight of a climbing backpack should not surpass 10% of the child’s body-weight so as not to hinder the growing skeletal system [18] (Fig. 14.7).

14.4 Conclusion

Climbing represents a sport which is beneficial for the health and development of children and adolescents (Fig. 14.8). Besides training, the skeletal musculature also has a cardiopulmonary effect, though to a lesser extent. The intensity and volume of the training should be adapted to the biological age of each athlete. High-intensity training modalities with added weights should be avoided before the end of puberty.



Fig. 14.5 Climbing shoes in children should never be too tight. Pressure marks as seen in this 6-year-old must be avoided



Fig. 14.6 Stress fracture—osteonecrosis of the bunion of a 16-year-old climber



Fig. 14.7 Children have no subjective risk assessment—it is therefore the responsibility of the person in charge to assure the safety in case of a fall. What may look spectacular in this picture is in reality sufficiently assured with mats and spotters



Fig. 14.8 Children enjoy bouldering. (Photo: Kilian Reil)

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